Signals 2/8/23

I Linear Transformations

Def: Let X and Y ber vector spaces. A transformation $L: X \to Y$ is linear if $\forall\, x_1, x_2 \in X$

- $L(\alpha x_1) = \alpha L(x_1)$
- $L(x_1 + x_2) = L(x_1) + L(x_2)$

A) Example

$X = C[0, \infty)$

$x(t) \in X$

$$L(x(t)) = \int_0^t H(\tau)\, x(t-\tau)\, d\tau$$

$$x(t) \longrightarrow \boxed{H(t)} \longrightarrow x(t) * H(t)$$

$X = C(-\infty, \infty)$; $Y = X$

$$T_{T_0}\, x(t) = \begin{cases} x(t) & |t| \le T_0 \\ 0 & \text{otherwise} \end{cases}$$

$X = \{$ Fourier transformable functions $\}$.
$Y = X$

$F : X \to Y$

$$Fx(t) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} \, dt$$

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$B : X \to Y$

X, Y same as this example

$$B_{B_0} = F^{-1} I_{B_0} F \, x(t)$$

In words: convert to frequency domain, truncate, convert back to time domain. In other words an ideal high pass filter.

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$X = \mathbb{R}^3 \quad Y = \mathbb{R}^2$

$L : X \to Y$

$$L(x_1, x_2, x_3) = (x_1 + 2x_2, \; 3x_2 + 4x_3)$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad L = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 4 \end{bmatrix}$$

$$L(\cdot) = L\underline{x}$$

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Def: Let $L: X \to Y$

The range of L, denoted by $R(L)$ is

$$R(L) = \{ y \in Y : y \in Lx,\ x \in X \} \subset Y$$

Def: The nullspace of L, denoted $N(L)$ is

$$N(L) = \{ x \in X : Lx = 0 \}$$

This is also known as the kernel.

B) Examples

$A = \mathbb{R}^{m \times n}$ $\quad A = [\underline{P}_1 \ \cdots \ \underline{P}_n]$ $\quad P_n \in \mathbb{R}^m$

$$Ax = x_1 \underline{P}_1 + x_2 P_2 + \cdots + x_n P_n$$

$A: \mathbb{R}^n \to \mathbb{R}^n$

$$R(A) = \underbrace{\text{span}(\{ \underline{P}_1, \cdots, \underline{P}_n \})}_{\text{Column space of } A}$$

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Suppose $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ $\quad R(A) = \text{span}\left(\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \right)$

2 dimensional range

$N(A) = \text{span}\left(\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}\right)$ - 1 dimensional NULL space

$A = \begin{bmatrix} q_1^T \\ \vdots \\ q_m^T \end{bmatrix}$ $\quad y^T A =$ linear combination of A

span of rows of A = row space = $R(A^T)$

$N(A^T) = \{y : y^T A = 0\}$ = Left NULL space of A

$L : C(-\infty, \infty) \to C(-\infty, \infty)$

$$L x(t) = \int_0^\infty x(\tau) H(t-\tau)\, d\tau$$

$$F\{L x(t)\} = F\{x(t)\} \cdot F\{H(t)\}$$

$$= X(f) \cdot H(f)$$

c) Projections and orthogonal Projections

Def: If V_1 and V_2 are subspaces of a vector space V

$$V_1 + V_2 = \{x = v_1 + v_2 : v_1 \in V_1,\ v_2 \in V_2\}$$

Def: V_1 and V_2 are subspaces of V are said to be disjoint if

$$V_1 \cap V_2 = \{0\}$$

Lemma: Let V_1 and V_2 be subspaces of vector space V. Then $\forall\, x \in V_1 + V_2$, there is a unique vector $v_1 \in V_1$ and $v_2 \in V_2$ s.t.

$$x = v_1 + v_2$$

iff V_1 and V_2 are disjoint.

Let V_1 and V_2 be disjoint vector spaces that are subspaces of V

$$x = v_1 + v_2$$

Define $P : V \to V_1$ by $P\underline{x} = v_1$ whenever $x = v_1 + v_2$

↳ This means given a vector $x = v_1 + v_2$, the operator P will return v_1.

Def: A linear transformation $P : V \to V$ is a projection if $P^2 = P$.

1) Example

$$T_{T_0}\, x(t) = \begin{cases} x(t) & -T_0 \le x \le T_0 \\ 0 & \text{otherwise} \end{cases}$$

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Theorem: Let P be a projection operator. The range of P and the nullspace of P are disjoint subspaces.

$$X = R(P) + N(P)$$

$\uparrow$

Entire space

If P is a projection operator, then $I - P$ is a projection operator.